Subject: Dorian Arnold's 2026 – 2027 Sabbatical Leave

To Whom It May Concern:

I seek an institutional host for a Sabbatical Leave from Summer 2026 to Summer 2027[1]. I have almost three decades of R&D experience in HPC Distributed, Runtime, and Operating Systems. My area of expertise includes Software Development, Debugging and Performance Analysis, Testing and Validation, Resource Management and Scheduling, Fault-tolerance, Communication Infrastructures, and Monitoring/Telemetry. Generally, we design and develop software infrastructures to improve both what is done and how efficiently – at massive scales of intra- and inter-node concurrency.

Broadly, my sabbatical goals are to study compelling opportunities for Systems and AI integration. Via specific case studies, I wish to refresh my field expertise and to deepen my knowledge of AI- and data-driven systems innovations (aka *AI for Systems*). I also am interested in system innovations to advance AI (aka *Systems for AI*). Target outcomes include high-impact research publications, useful software or data artifacts, and sustained partnerships.

My Applied AI interests include Transformer Models (large/small language and multimodal), Reasoning Models, and Agentic AI. After my sabbatical, I should understand which methods best-suit which applications. But also, I should have applied knowledge of at least one of these areas for targeted problems. Potential topics include domain-specific copilots or autonomous agents applied within the HPC ecosystem, for example, in my expertise areas. Even before this AI resurgence, my work aims to simplify high-performance computing by masking system complexity with intuitive, high-level abstractions, while admitting performance and scalability. I am excited for how AI can advance this goal.

Summarily, I seek to adapt my expertise by integrating new AI methods to revisit recurrent or solve new computer systems challenges – a fruitful sabbatical experience that equally will reward my host and my own professional growth.

Thank you for your time and consideration, and I welcome further discussion to explore the possibilities.

Sincerely,

Dorian Arnold
Associate Professor, Computer Science Department, Emory University

Attached: 2-Page Résumé

[1]Duration up to 15 months – as early as mid-May 2026 to as late as mid-August 2027.

Dorian Cecil Arnold, Ph.D.

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Technical Interests: Computer System Software R&D, including AI- and data-driven approaches.
Other Interests: University Leadership; Computing and STEM Education; Inclusive Workforce Development.

Scientist, Architect, Educator. Manager, Administrator, Leader!

30 Years	Distributed, Runtime, and Operating Systems Software R&D, including HPC Systems, Fault-tolerance, Debugging and Performance Analysis, and Middleware.
18+ Years	Small and large team Management with many direct reports; Developing and executing strategic vision; Developing and managing multi-million dollar budgets, including six and seven figure contracts.
8 Years	Directing a complex multi-department graduate program; Overseeing 200% and 100% growth in students and faculty, respectively; Student Recruitment, Admission and Progress; Curricular Revisions; and Professional Development.

Technical Highlights and Proficiencies

70+	refereed research papers and one patent
2400+	citations. h-index: 26; i40-index: 13; i10-index: 41
2	R&D Top 100 awards
$13M	of collaborative sponsored research, including $3.4M of PI funding
Partnerships	strong industry, DOE lab, and university collaborations, including $1.6M corporate funding
Software	production software packages deployed on leadership platforms across the world
Renown	Distinguished ACM Speaker (2017-2020); IEEE Senior Member; many Invited Keynote/Plenary Talks
★★★★	C/C++, Linux/Unix, Performance Tools, Telemetry, Job Schedulers, Resource Managers
★★★☆	Java, Python, MPI, Debuggers, Software Testing, Parallel File Systems, Computer System Simulation
★★☆☆	OpenMP, Matlab, Fortran, Scientific and Numerical Libraries, Scrum, Agile Development
★☆☆☆	Large Language Models, Kubernetes/Docker, Workflow Orchestration

Professional Appointments

2017 –	**Associate Professor**, *Computer Science*, Emory University
2018 –	**Director of Graduate Studies**, *Computer Science and Informatics*, Emory University
2009 – 2017	**Assistant/Associate Professor**, *Computer Science*, University of New Mexico
2013	**Summer Faculty**, *Department of Scalable System Software*, Sandia National Labs
2011 – 2013	**Affiliate Research Scientist**, *Ultrascale Systems Research Center*, New Mexico Consortium
2009 – 2010	**Visiting Scientist**, *Ctr. Applied Scientific Computing*, Lawrence Livermore National Lab
2006	**Technical Scholar**, *Ctr. Applied Scientific Computing*, Lawrence Livermore National Lab
1999 – 2001	**Research Associate**, *Innovative Computing Laboratory*, University of Tennessee

Select Projects and Project Supervision

Software Synthesis	Statistical and AI-based methods for synthesizing *Application Analogs*, proxies, benchmarks, traces, or other workload representatives of real applications. (Ongoing SNL collaboration.)
Telemetry	Practical monitoring optimizations, via data-driven principles for configuring spatial (which sensors) and temporal (sampling frequency) monitoring and data reductions. (Ongoing TACC collaboration.)

Resilience: Studied various aspects of fault-tolerance, primarily using software prototypes and simulation to evaluate checkpoint/restart protocols under various configurations, workloads, and platform environments. This project resulted in many high-impact research publications and the training of multiple PhD, MS, and BS students. (Past collaboration with Sandia National Labs and University of Tennessee.)

Software Testing: Managed team of software engineers and students in the design and development of production software regression and specification compliance tests for Cray Programming Environment suite, including C, C++ and Fortran tests for Cray Compilers, MPI, OpenMP, SHMEM, and CUDA. (2014 – 2020, supported by Hewlett Packard Enterprise, formerly Cray Inc.)

Overlay Networks: Founding researcher/developer of the multicast/reduction network, a tree-based overlay network for flexible, reliable data multicast and aggregation. MRNet has improved by orders of magnitude the scalability of many tools and applications and is used at laboratories and universities worldwide.

Scalable Debugging: Founding researcher/developer of the stack trace analysis tool (STAT), which uses MRNet to organize application stack traces into spatiotemporal call graph prefix trees for scalable debugging. STAT won an R&D Magazine Top 100 award in 2011 and remains in production use on large HPC systems today.

Grid Computing: Technical lead and developer of the network-enabled solvers (NetSolve), a client-server middleware for remote access to complex hardware and software resources. NetSolve has been used in many real applications across many domains and won an R&D Magazine Top 100 award, in 2000.

Rollback Recovery: Developed the Checkpoint Library for Unix Based Systems (CLUBS), for user-transparent checkpoint/restart, integrating optimizations like efficient file formats and exploiting copy-on-write semantics.

PhD Theses:
- W. Schonbein (2020), "Intelligent Networks for High Performance Computing"
- S. Gutiérrez (2018), "Adaptive Parallelism for Coupled, Multithreaded Message-Passing Programs"
- D. Ibtesham (2017), "Improving Large Scale Application Performance via Data Movement Reduction"
- T. Groves (2017), "Characterizing and Improving Power and Performance in HPC Networks"

MS Theses:
- Y. Wang (2016), "Generating Traces of Application Behavior Using Generative Adversarial Networks"
- B. Klemme (2016), "Software Testing of Parallel Programming Frameworks"
- L. Frey (2014), "Introducing Modularity into Complex Software Test Suite Frameworks"
- J. Goehner (2011), "Efficiently Bootstrapping Extreme Scale Software Systems"

BS Theses:
- M. Lafrance (2026), "On Checkpoint/Restart Viability for Exascale High Performance Computing"
- C. Chance (2022), "Zoom Audio Transcription Accuracy for African American Vernacular English"
- H. Feng (2022), "PADL: Persistent Application Data Library"
- Y. Jiang (2022), "Synthetic Generation of Datasets With Complex Attributes"

Select Impactful Papers

[SC11] Ferreira et al and Arnold. Evaluating ... Process Replication Reliability for Exascale Systems.

[SC08] Lee, Ahn, Arnold et al. Lessons Learned at 208k: Towards Debugging Millions of Cores.

[IPDPS07] Arnold et al. Stack Trace Analysis for Large Scale Applications.

[SC03] Roth, Arnold, and Miller. MRNet: A Software-based Multicast/reduction Network for Scalable Tools.

[ICPP12] Ibtesham, Arnold et al. On ... Compression for Reducing the Overheads of Checkpoint/restart ...

Professional Preparation

2008 **Ph.D., Computer Sciences**, *University of Wisconsin*, Advisor: Dr. Barton Miller

1998 **M.S., Computer Science**, *University of Tennessee*, Advisor: Dr. James Plank

1996 **B.S. summa cum laude, Mathematics, Computer Science**, *Regis University*

1994 **A.S., Mathematics, Physics, Chemistry**, *St. John's Junior College (Belize)*